



MANIPAL ACADEMY OF HIGHER EDUCATION

DEPARTMENT OF MATHEMATICS
IV SEMESTER B.TECH. (Mathematics and Computing)
MAKE UP EXAMINATION JUNE 2025
13-06-2025 2:00pm-5:00pm

VECTOR ANALYSIS AND COMPLEX VARIABLES [MAT 2238]

Marks: 50

Duration: 180 mins.

DESCRIPTIVE

Answer all the questions.

Section Duration: 180 mins

Any missing data may be assumed suitably

- 1A) The temperature of a thin plate is described by a scalar field f , the temperature at (x, y) being $f(x, y)$. Rewriting in terms of polar co-ordinates $x = r \cos(\theta)$ and $y = r \sin(\theta)$, the temperature becomes a function of r and θ , given by $\varphi(r, \theta) = f(r \cos(\theta), r \sin(\theta))$. Express the first order partial derivatives $\frac{\partial \varphi}{\partial r}$ and $\frac{\partial \varphi}{\partial \theta}$ in terms of the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$. Hint: Use Chain Rule. (4)
- 1B) Find the constants a, b, c so that the vector $\vec{v}(x, y, z) = (-4x - 3y + az)\hat{i} + (bx + 3y + 5z)\hat{j} + (4x + cy + 3z)\hat{k}$ is irrotational. (3)
- 1C) Find the normal vector and the equation for the tangent plane to the surface $xz^2 + xy^2 = z - 1$ at the point $(1, -3, 2)$. (3)
- 2A) Compute the directional derivative of $f(x) = \|x\|^2$ at $\vec{a} = (1, 1, 0)$ in the direction of $\vec{b} = (1, 2, 2)$. Here, $x \in \mathbb{R}^3$. (3)
- 2B) Find all the complex cube roots of -1 . (3)
- 2C) Let $z = x + iy$. If the complex valued function $f(z) = u(x, y) + iv(x, y)$ and its conjugate $\bar{f}(z) = u(x, y) - iv(x, y)$ are both analytic in a non-empty domain D (domain means open subset of $\mathbb{C}$), show that $f(z)$ must be a constant throughout D . Hint: Use (4)

Cauchy Riemann equations on f and $\bar{f}$.

3A) Show that $u(x,y) = y^3 - 3x^2y$ is harmonic on $\mathbb{R}^2$ and find a harmonic conjugate of $u(x,y)$. (3)

3B) Find all z for which $e^z = 1 + \sqrt{3}i$. (4)

3C) Evaluate the integral (in counterclockwise sense) $\oint_C \frac{1}{4z-1} dz$ where C is the unit circle. (3)

4A) Let C denote the line segment from $z=0$ to $z=1+i$. Find an upper bound for the absolute value of the integral $\oint_C z^2 dz$. (3)

4B) Evaluate the following contour integral in the counterclockwise sense where C is the circle $|z+i|=1$. (4)

$$\oint_C \frac{(z^4 - 3z^2 + 6)}{(z+i)^3} dz$$

4C) Evaluate the contour integral: (3)

$$\oint_C \frac{\sin(z)}{(z^2 + \pi^2)} dz,$$

where the contour C is the circle $|z|=2$ traversed counterclockwise.

5A) If a vector valued function $\vec{f}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by $\vec{f}(x,y,z) = x\hat{i} + y\hat{j} + z\hat{k}$, then determine the Jacobian matrix of $\vec{f}$. (3)

5B) Let $K=f(A)=A^2$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a 2 x 2 matrix with real entries. Let the $(p,q)^{th}$ component of K be denoted $K_{p,q}$. Find $\frac{\partial K_{1,1}}{\partial a}$, $\frac{\partial K_{1,2}}{\partial b}$ and $\frac{\partial K_{2,2}}{\partial d}$. (3)

5C) Consider a simple neural network with an input layer of 2 neurons ($x_1 = 1.5, x_2 = 2.0$) and an output layer with 1 neuron (target output $y = 3.0$) activated by the sigmoid function $\sigma(x) = (1 + e^{-x})^{-1}$. The cost/loss function is the mean square error (MSE) $C(a^1) = \frac{1}{2}(a^1 - y)^2$, and the learning rate α is 0.5, and the initial parameters are (4)

$w_1 = -1.2$, $w_2 = 0.8$, $b = 0.5$. Perform one step each of forward propagation and backward propagation using gradient descent. Present all computed values (including intermediate variable z , all gradients and updated parameter values) in a table, rounding all values to 4 decimal places.

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